



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 3310

Digital Fundamentals and System Design

Wenchao Miao

Outline Lecture 6 Finite State Machines and Counters



THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

- 1 Learning Outcomes**
- 2 FSM Fundamentals**
- 3 Counter Design Using FSM**
- 4 Integrated Circuit (IC) Counters**
- 5 JK Flip-Flop**
- 6 Applications of Counters**

1. Learning Outcomes

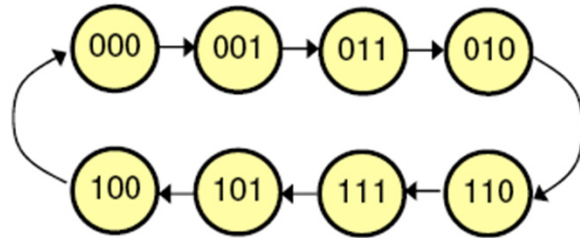
- 1. Describe finite state machine (FSM) and FSM formalism**
- 2. Describe synchronous counters**
- 3. Design synchronous counter circuits using FSM formalism**
- 4. Describe Integrated Circuit (IC) counters**
- 5. Recognize the difference between 4-bit binary and decade counters**
- 6. Describe synchronous cascading**
- 7. Determine/modify the modulus of a synchronous counters**
- 8. Use cascaded counters to achieve a higher modulus**
- 9. Describe the operation of JK flip flop and SR latch**
- 10. Explain a how digital clock operates**

Main Content

Finite State Machines and Counter Basics

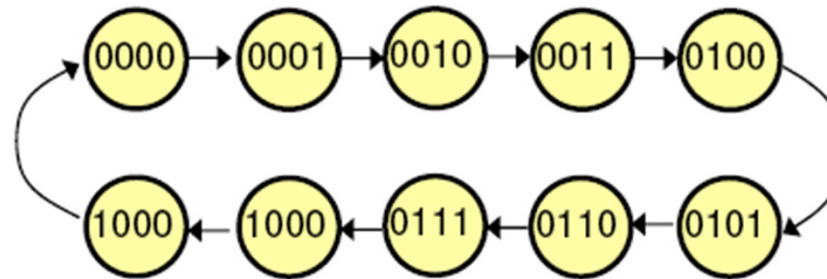
- **FSM as Sequential Logic Formalism**
- **Modulus in Counters (Number of States)**
- **Counter Types: Asynchronous vs Synchronous**

- For combinational logic, BOOLEAN equation(s) or a TRUTH table that leads to Boolean equation(s) are the *mathematical formalism*
- **Mathematical formalism:** a description of a system (or circuit) in formal mathematical or logical terms.
- For sequential logic, Boolean equation(s) or Truth tables are **NOT** sufficient!
- **For sequential logic, Finite State Machines (FSMs) are the mathematical formalism.**
- **FSM also refers to a SEQUENTIAL circuit having a limited (FINITE) number of states occurring in a prescribed order.**
- **Counters and registers are a type of FSM.**

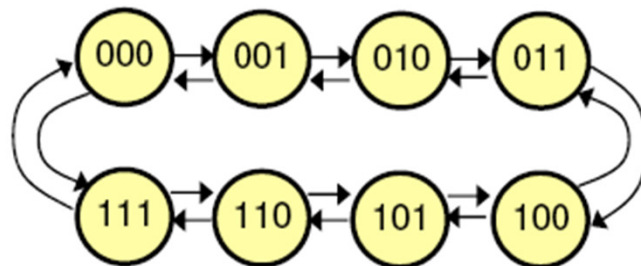


gray-code counter
3 FFs required

The number of states in a binary counter is called **MODULUS**.



BCD counter
MODULUS – 10
4 FFs required



up/down counter...

MODULUS – 8
3 FFs required

2 counter types based on how FFs are clocked

Asynchronous (Ripple) Counters: the first FF (LSB) is clocked by the external clock pulse and then each successive FF is clocked by the output of the preceding FF.

Synchronous Counters: the clock input is connected to all of the FFs so that they are clocked simultaneously (all synchronized).

We will first study Synchronous counters

Main Content

Designing Synchronous Counters with FSM

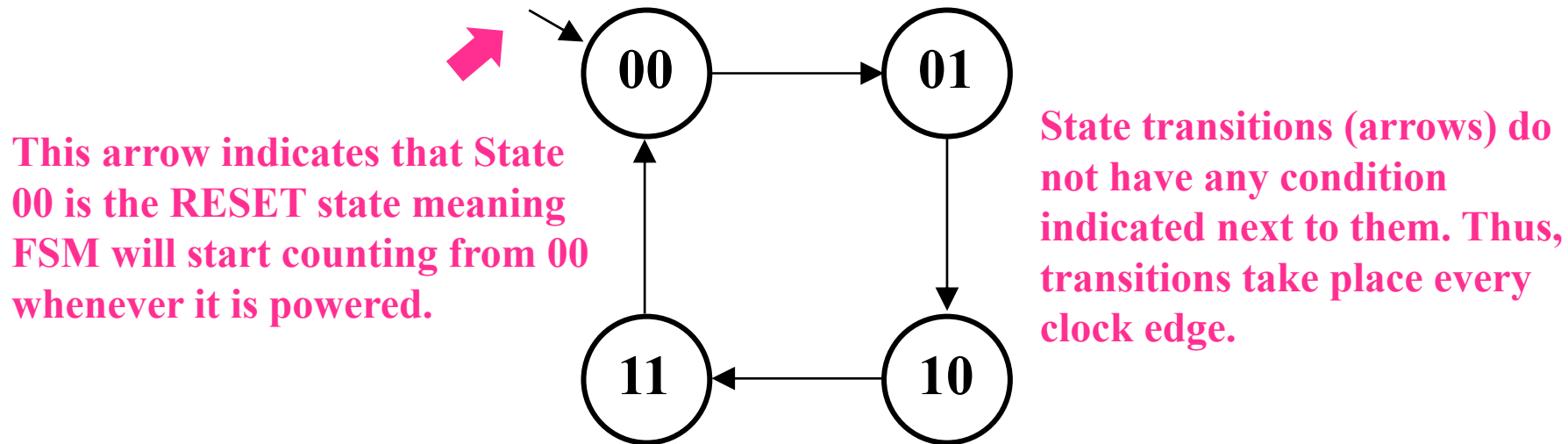
- Steps for Counter Design
- Example I: 2-Bit Binary Up Counter (No Input/Output)
- Example II: 2-Bit Binary Up Counter with Enable Input

- 1. Obtain specification of the desired counter**
- 2. Create a state diagram from specification.**
- 3. Create a state table from state diagram.**
- 4. Derive the next-state logic and output logic (if any) expressions.**
- 5. Implement circuit described by expressions.**

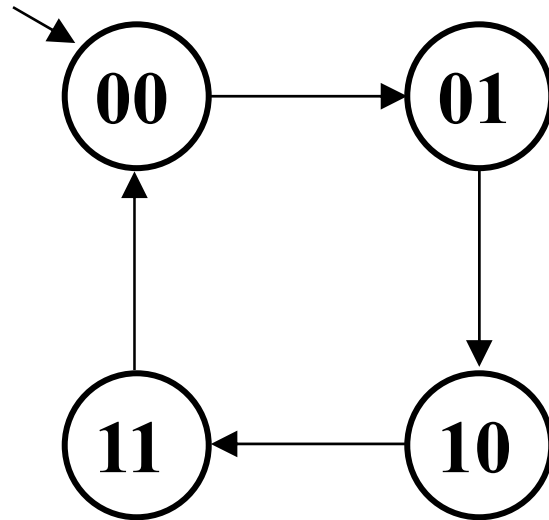
1. Obtain specification of the desired circuit.

- Design a 2-bit binary up counter
- Counter has no input/output (just receives a clock signal)

2. Create a state diagram from specification.



3. Create a state table from state diagram.



Q_i : present state
 Q_i^+ : next state

Present State	Next State
Q_1Q_0	$Q_1^+Q_0^+$
0 0	0 1
0 1	1 0
1 0	1 1
1 1	0 0

4. Derive the next-state logic expressions.

Treat it as a 2-input TRUTH table
Inputs: Q_1, Q_0 ; Outputs: Q_1^+, Q_0^+
Express outputs in terms of inputs

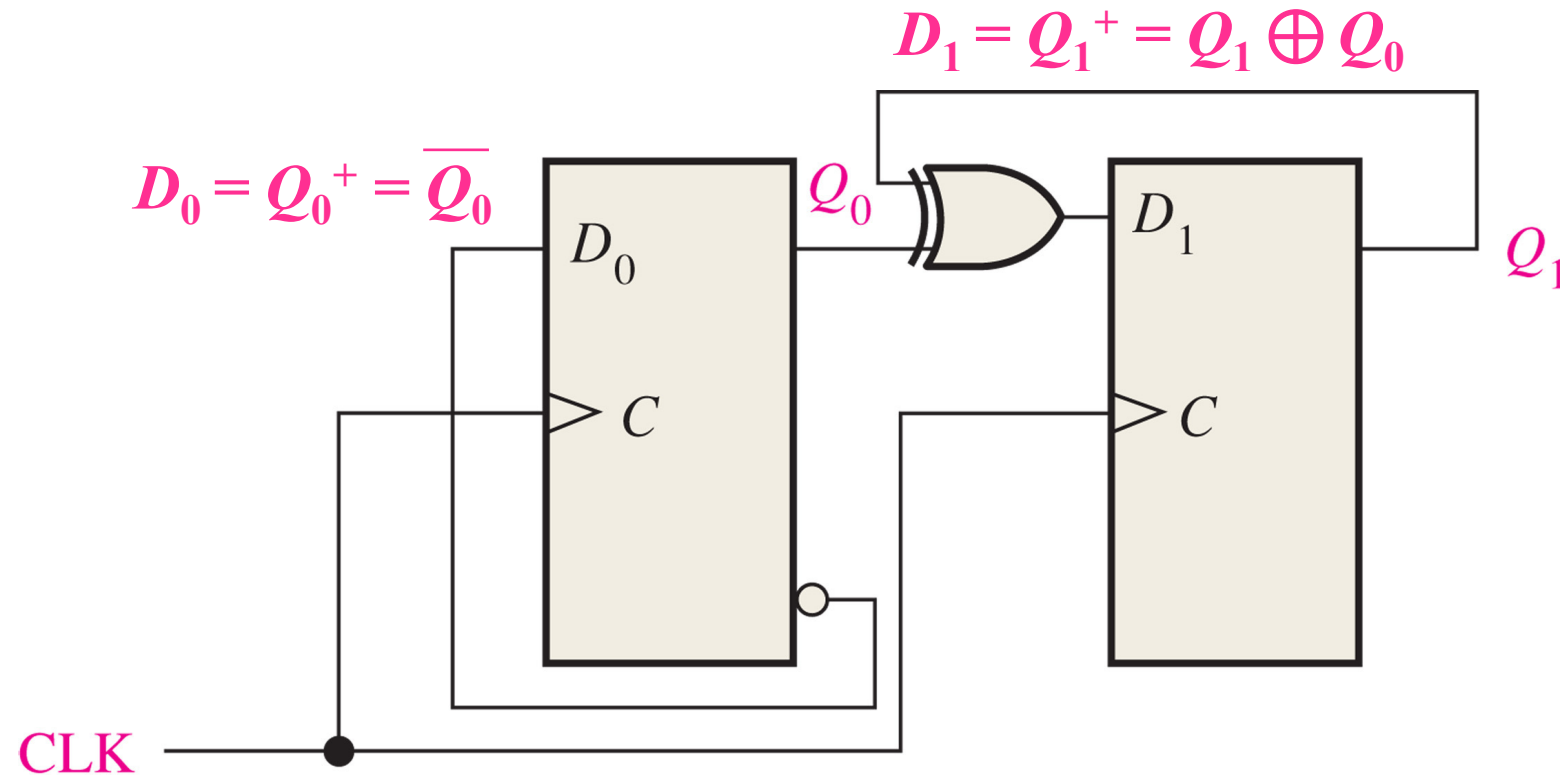
Present State	Next State
$Q_1 Q_0$	$Q_1^+ Q_0^+$
0 0	0 1
0 1	1 0
1 0	1 1
1 1	0 0

Remember for D flip flop
Next State is the D input $Q_i^+ = D_i$
No need to create extra columns for D_i

$$D_1 = Q_1^+ = Q_1 \oplus Q_0$$

$$\begin{aligned} D_0 = Q_0^+ &= \overline{Q_1} \overline{Q_0} + Q_1 \overline{Q_0} \\ &= \overline{Q_0} (\overline{Q_1} + Q_1) \\ &= \overline{Q_0} \underbrace{1}_{\text{pink 1}} \end{aligned}$$

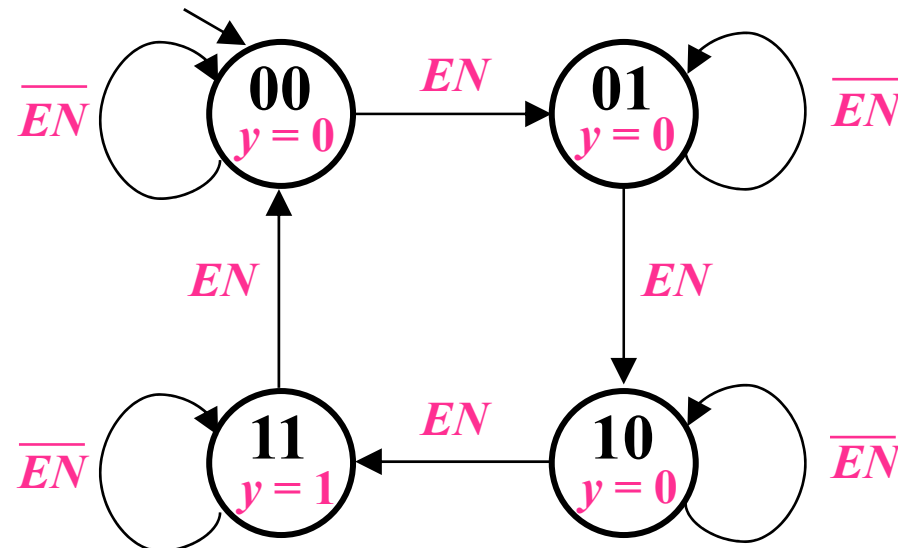
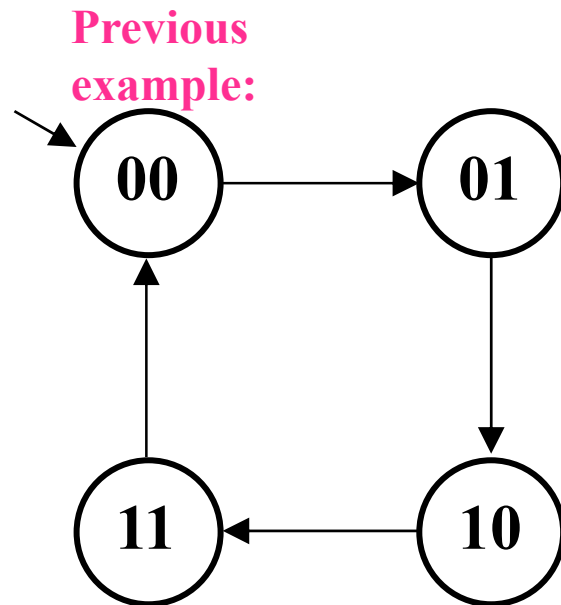
5. Implement the circuit described by logic.



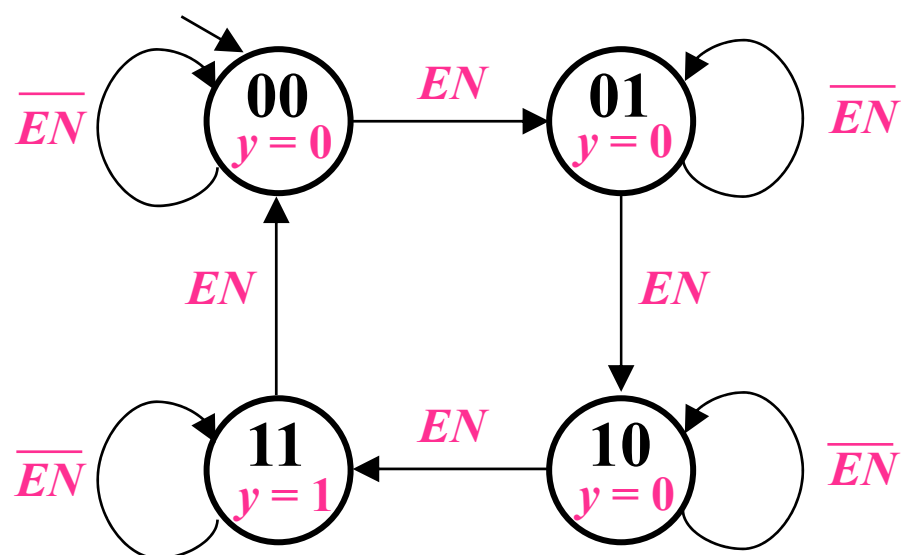
1. Obtain specification of the desired circuit.

- Design a 2-bit binary up counter
- Counter has one input and one output
- Input: EN ($EN = 1$ keep counting; $EN = 0$ stop counting)
- Output y ($y = 1$ counter has reached the final count value)

2. Create a state diagram from specification.



3. Create a state table from state diagram.



$y = 1$ when Q_1Q_0 is 11, otherwise $y = 0$

Q_i : present state, Q_i^+ : next state

Input	Present State	Next State	Output
EN	Q_1Q_0	$Q_1^+Q_0^+$	y
0	Q_1Q_0	Q_1Q_0	0/1
1	0 0	0 1	0
1	0 1	1 0	0
1	1 0	1 1	0
1	1 1	0 0	1

4. Derive the next-state logic expressions.

Treat it as a 3-input TRUTH table

Inputs: EN, Q_1, Q_0

Outputs: Q_1^+, Q_0^+, y

Note: $Q_i^+ = D_i$

Input	Present State	Next State	Output
EN	$Q_1 Q_0$	$Q_1^+ Q_0^+$	y
0	$Q_1 Q_0$	$Q_1 Q_0$	0/1
1	0 0	0 1	0
1	0 1	1 0	0
1	1 0	1 1	0
1	1 1	0 0	1

Express the outputs in terms of inputs

When $EN = 1$ D_1 is Q_1 XOR Q_0 ; D_0 is $\overline{Q_0}$

When $EN = 0$ D_1 follows Q_1 ; D_0 follows Q_0

$$D_1 = Q_1^+ = EN (Q_1 \oplus Q_0) + \overline{EN} Q_1$$

Proof
next slide $= (EN Q_0) \oplus Q_1$

$$D_0 = Q_0^+ = EN \overline{Q_0} + \overline{EN} Q_0$$

$$= EN \oplus Q_0$$

$$y = EN Q_1 Q_0 + \overline{EN} Q_1 Q_0$$

$$= (EN + \overline{EN}) Q_1 Q_0$$

$$= Q_1 Q_0$$

$$D_1 = Q_1^+ = EN(Q_1 \oplus Q_0) + \overline{EN} Q_1$$

Boolean manipulation
to show equivalence

$$= (EN Q_0) \oplus Q_1$$

$\overline{EN} Q_1(Q_0 + \overline{Q_0})$

Expand the first equation:

$$D_1 = Q_1^+ = EN \overline{Q_1} Q_0 + \overline{EN} Q_1 Q_0 + \overline{EN} Q_1 \overline{Q_0} + EN Q_1 \overline{Q_0}$$

Repeat the third term (this is permitted in Boolean algebra):

$$= EN \overline{Q_1} Q_0 + \overline{EN} Q_1 Q_0 + \overline{EN} Q_1 \overline{Q_0} + \overline{EN} Q_1 \overline{Q_0} + EN Q_1 \overline{Q_0} + EN Q_1 \overline{Q_0}$$

Pair the colored terms in common parentheses:

$$= EN \overline{Q_1} Q_0 + \overline{EN} Q_1 (Q_0 + \overline{Q_0}) + (\overline{EN} + EN) Q_1 \overline{Q_0}$$

$$= EN \overline{Q_1} Q_0 + \overline{EN} Q_1 + Q_1 \overline{Q_0}$$

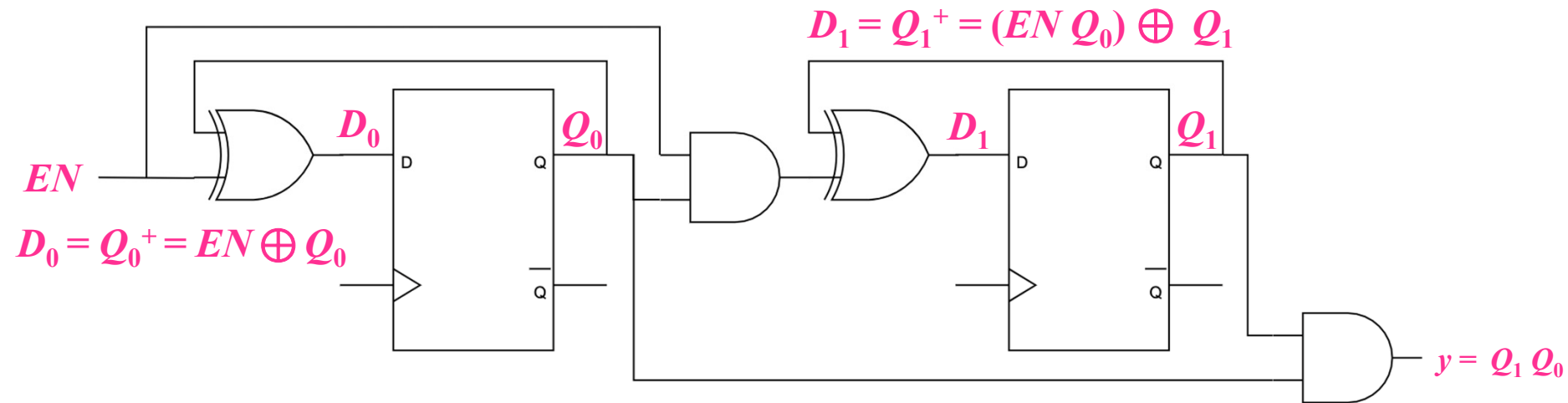
$$= EN \overline{Q_1} Q_0 + Q_1 (\overline{EN} + \overline{Q_0})$$

Apply de Morgan Law

$$= EN \overline{Q_1} Q_0 + Q_1 (\overline{EN Q_0})$$

$$= (EN Q_0) \oplus Q_1$$

5. Implement the circuit described by logic.

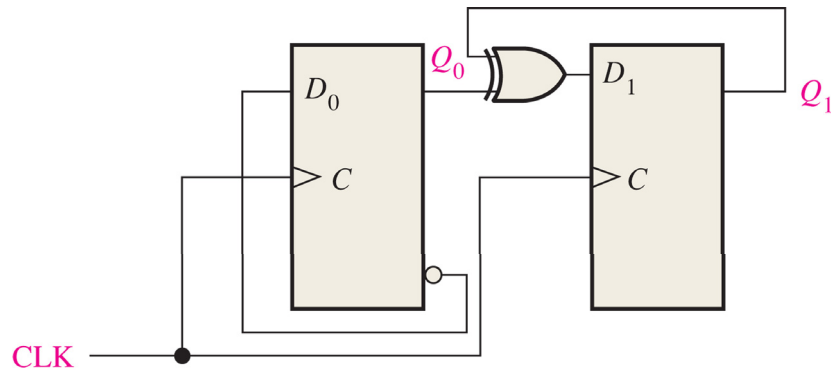


Clock connections are implicit meaning although not shown explicitly, we know both FF receives the same clock signal

For n-bit binary up counter, you can repeat red block as many bits as you need. The number of inputs to the output AND gate increases accordingly.

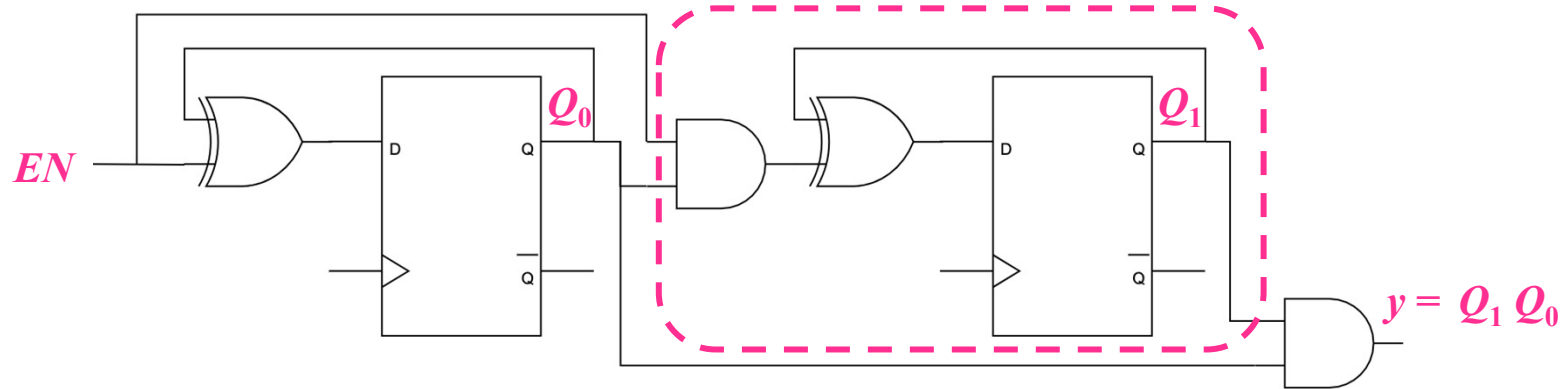

$$y = Q_1 Q_0$$

Synchronous vs Asynchronous Counters



Both counters are “Synchronous counters” because both FFs are triggered by the same clock signal (both change output at the same time synchronous with the clock).

There are also “Asynchronous counters” where each FF is triggered by a separate clock signal. We will see Asynchronous counters in the coming lectures



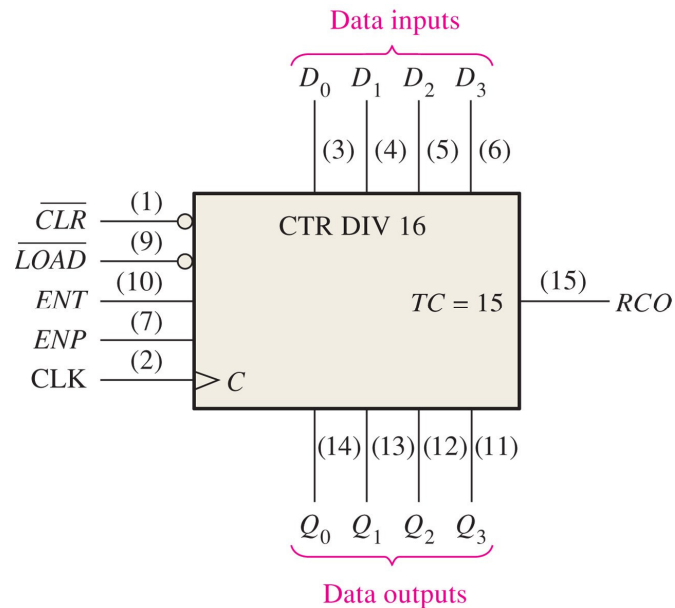
Main Content

Practical Implementation of Counters

- 74HC163 4-Bit Synchronous Binary Counter
- 74HC190 Up/Down Decade Counter
- Cascading Counters (Modulus Multiplication and Truncated Sequences)
- Digital Clock

4-bit Synchronous binary counter

The 74HC163 4-bit synchronous binary counter.
(The label CTR DIV 16 indicates a counter with sixteen states, modulus – 16)



The RCO output, in conjunction with the two enable inputs, allows these counters to be cascaded for higher count sequences.

The counter can be synchronously preset to any 4-bit binary number through parallel data inputs.

When the LOAD input is LOW, the counter will load the data inputs on the next clock pulse.

When the clear input (CLR) is LOW, the counter will clear the data output on the next clock pulse.

ENP and ENT are ENABLE inputs. Both must be HIGH for the counter to sequence through its binary states.

When at least one input is LOW, the counter is disabled.

The ripple clock output (RCO) goes HIGH when the counter reaches the last state in its sequence of fifteen, called the terminal count (TC = 15).

The inputs are synchronous inputs meaning, they take affect after the clock edge.

Timing Diagram



The 74HC163 4-bit synchronous binary counter

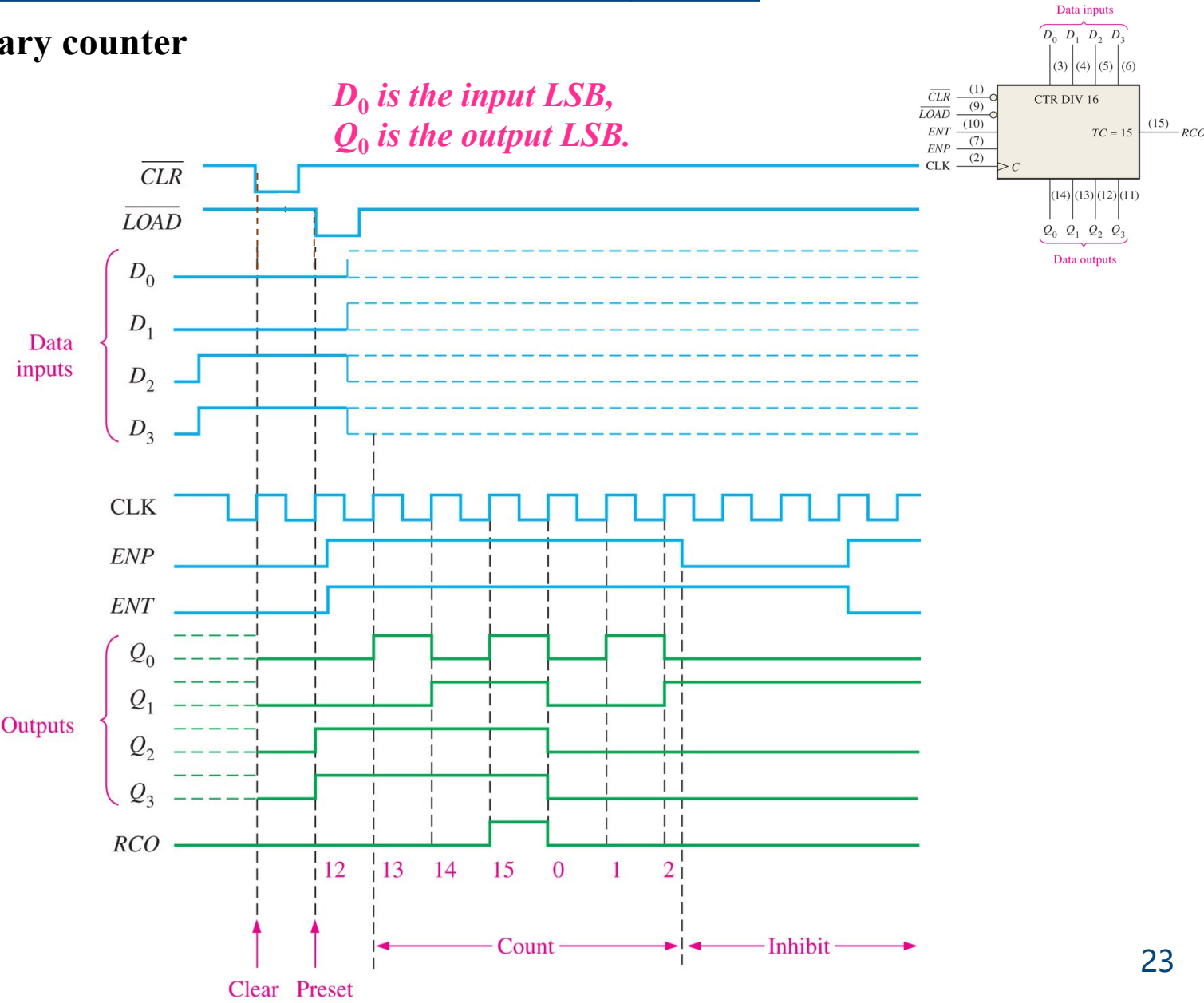
The LOW pulse on the CLR input causes all the outputs (Q_0 , Q_1 , Q_2 , and Q_3) to go LOW.

The LOW pulse on the LOAD input loads the data inputs (D_0 , D_1 , D_2 , and D_3). This is the PRESET operation.

The counter preset to 12 (1100) and then counting up to its terminal count, 15 (1111).

After 15 (1111), the counter cycles back to 0 (0000) and continues counting as long as both ENP and ENT are HIGH

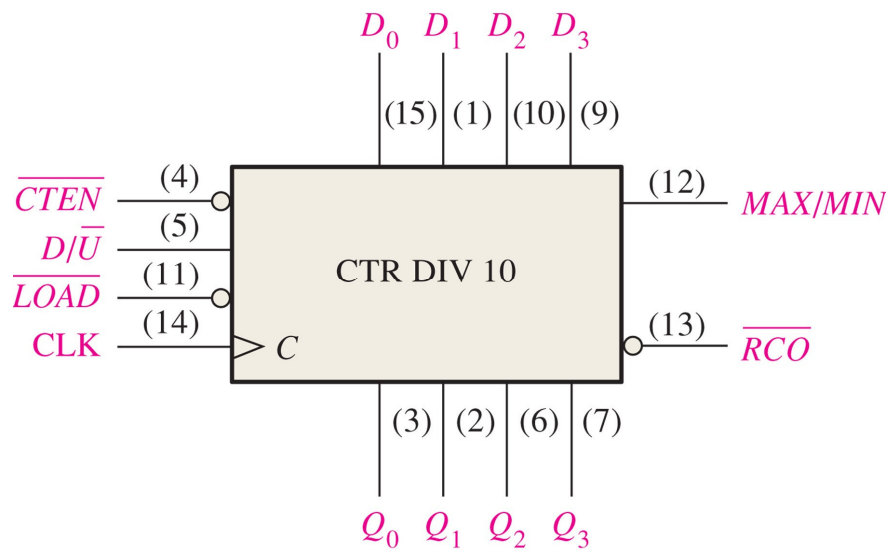
The counting is inhibited and pauses at state 0010 because first ENP and then ENT goes LOW.



Up/Down Synchronous decade counter

The 74HC190 up/down synchronous decade counter.

(The label CTR DIV 10 indicates a counter with 10 states, modulus – 10)



*The Up/down input (D/U) for count direction. **HIGH** input for count down; **LOW** input for up.*

*When the **LOAD** input is **LOW**, the counter loads the data inputs asynchronously (without **CLK** pulse).*

*When the count enable (**CTEN**) input is **HIGH**, the counter is inhibited.*

*The **MAX/MIN** output produces a **HIGH** pulse when the terminal count is reached: 9 (1001) in the **UP** mode 0 (0000) in the **DOWN** mode.*

*The **MAX/MIN** output, the ripple clock output (**RCO**), and the count enable input (**CTEN**) are used when cascading counters.*

Timing Diagram



The 74HC190 up/down synchronous decade counter

Note LOAD is an asynchronous input

The counter preset to 7 (0111) and then goes through a count-up sequence followed by a count-down sequence.

The LOW pulse on the LOAD input loads the data inputs (D_0 , D_1 , D_2 , and D_3). This is the PRESET operation.

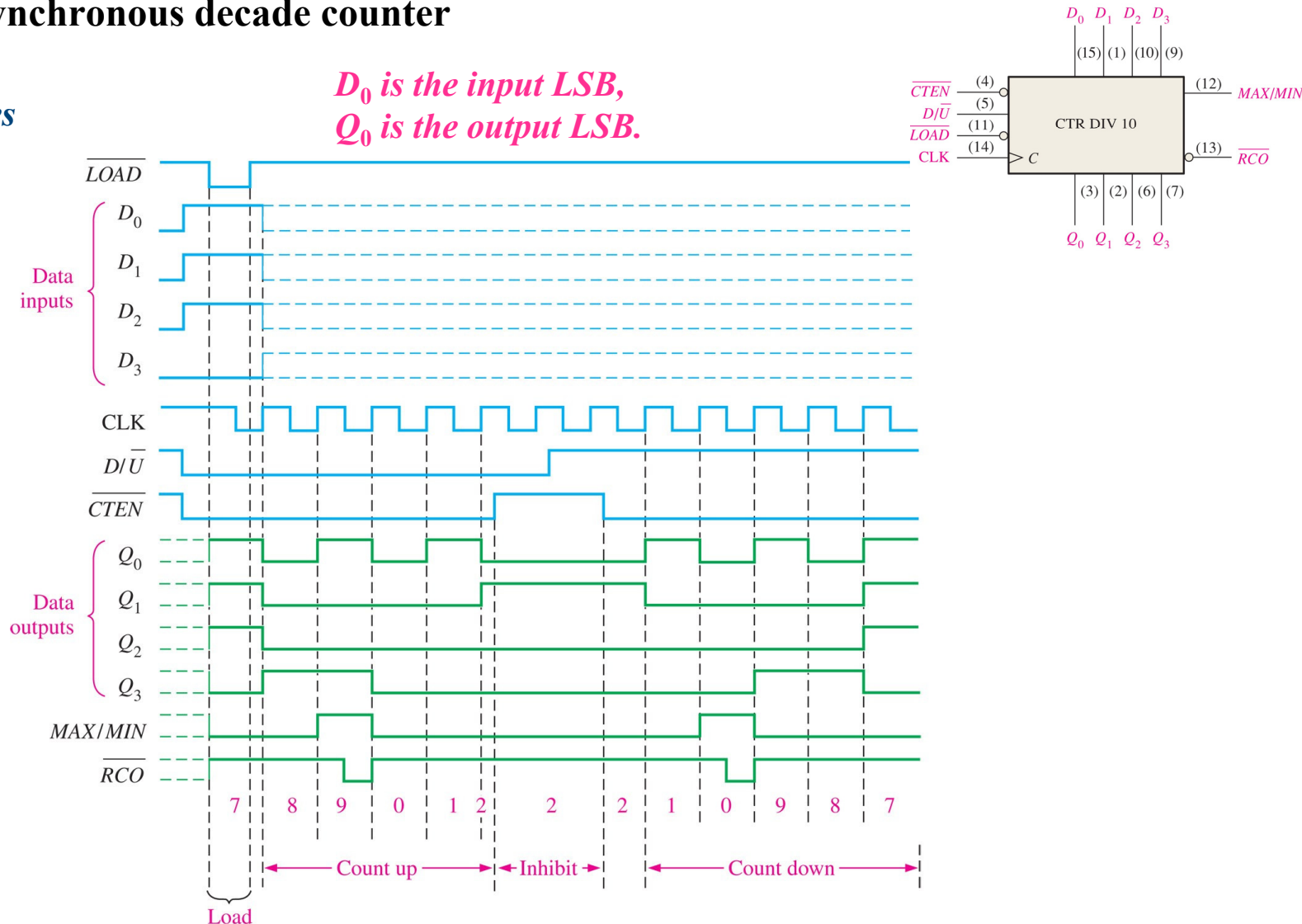
After 9 (1001), the counter cycles back to 0 (0000) and continues counting up as long as both CTEN and D/U inputs are LOW

The counting is inhibited and pauses at state 0010 because CTEN becomes HIGH.

The counting begins when CTEN becomes LOW but in the DOWN direction because D/U is HIGH now.

MAX/MIN becomes HIGH and RCO goes LOW when 1001 or 0000 being reached.

*D_0 is the input LSB,
 Q_0 is the output LSB.*

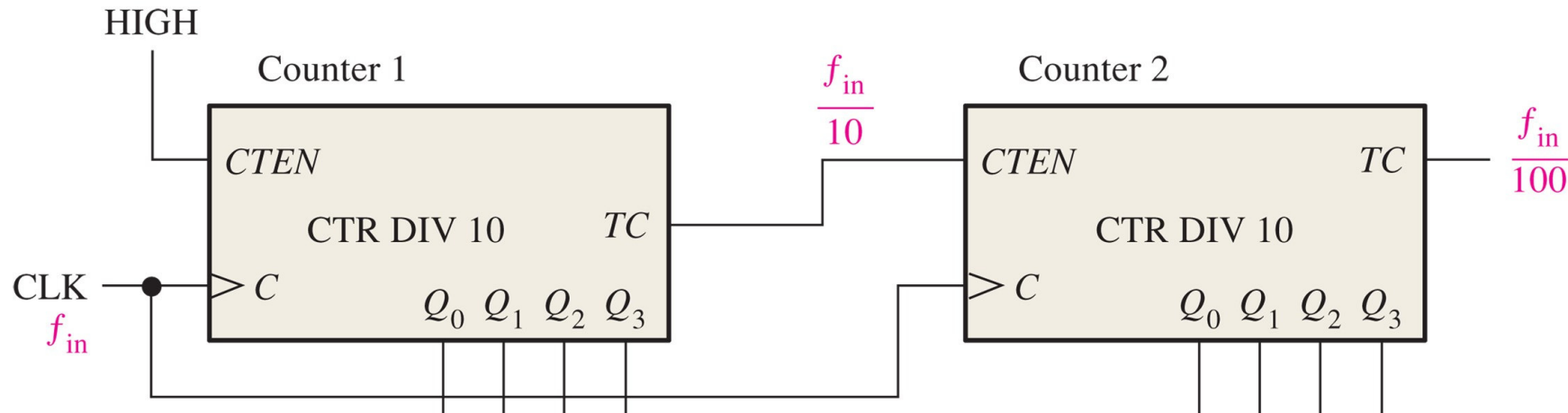


The RCO produces a low-level output pulse equal in width to the low-level portion of the clock input when an overflow or underflow condition exist

Synchronous Cascading

A modulus-100 counter using two cascaded decade counters.

Cascaded counters (sometimes called countdown chains) are often used to divide a high-frequency clock signal to obtain highly accurate pulse frequencies.



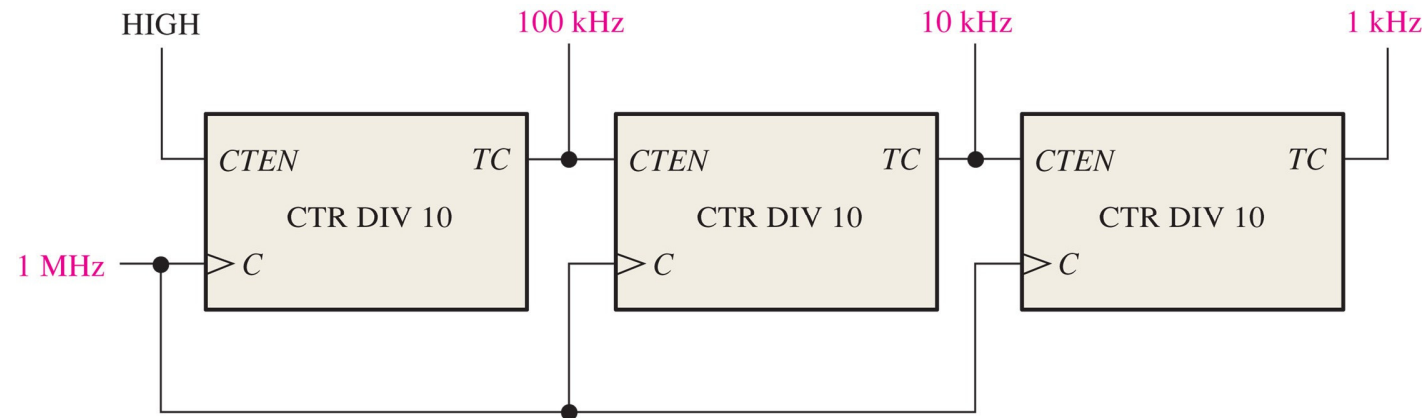
Counter 2 is inhibited by the LOW on its CTEN input until counter 1 reaches its terminal state (TC output goes HIGH). Counter 2 goes from its initial state to its second state during the first clock pulse (Clock cycle 10) after counter 1 TC HIGH

Counter 1 must go through ten complete cycles before counter 2 completes its first cycle i.e., for every ten cycles of counter 1, counter 2 goes through one cycle.

*Counter 2 will complete one cycle after one hundred clock pulses. The overall modulus of these two cascaded counters is $10 * 10 = 100$.*

Synchronous Cascading

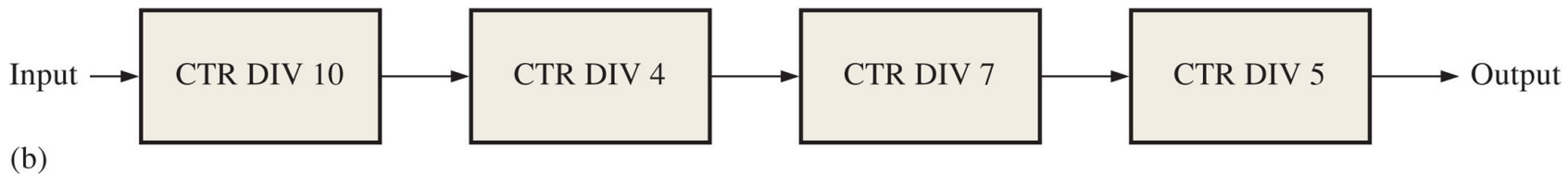
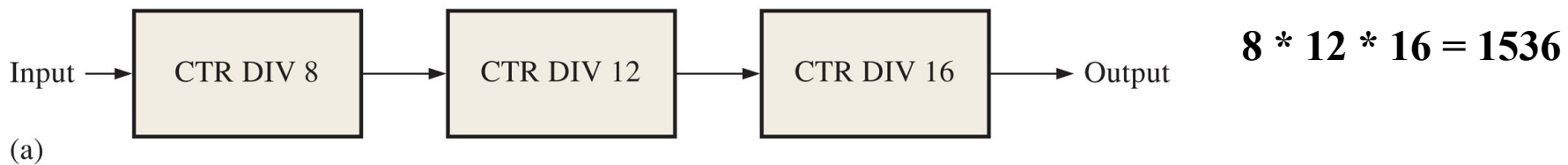
Three cascaded decade counters forming a divide-by-1000 frequency divider with intermediate divide-by-10 and divide-by-100 outputs.



The above “countdown chain” divides 1 MHz clock signal by 10 at three successive stages to obtain 100, 10 and 1 kHz signals at the counter TC output of the respective stage.

full-modulus cascading: overall modulus (divide-by-factor) that is the product of the individual moduli of all the cascaded counters.

Determine the overall modulus of the cascaded counter configurations

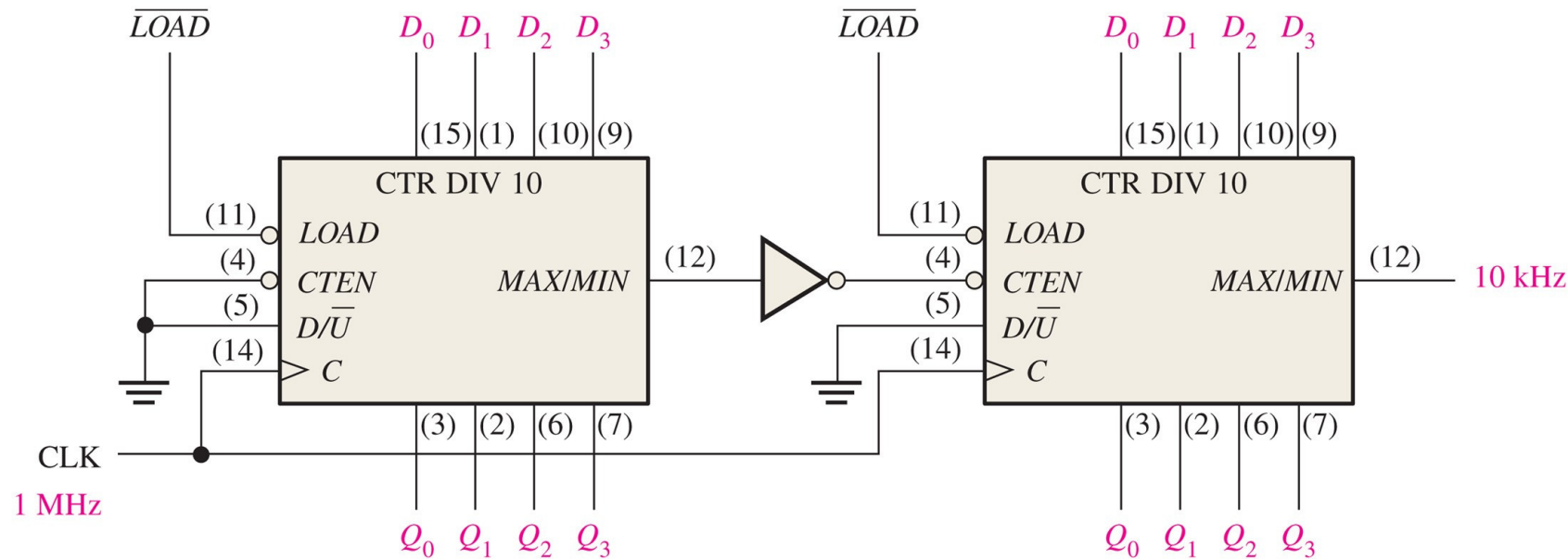


$$10 * 4 * 7 * 5 = 1400$$

Full-modulus Cascading



A divide-by-100 counter using two 74HC190 up/down decade counters connected for the up sequence.



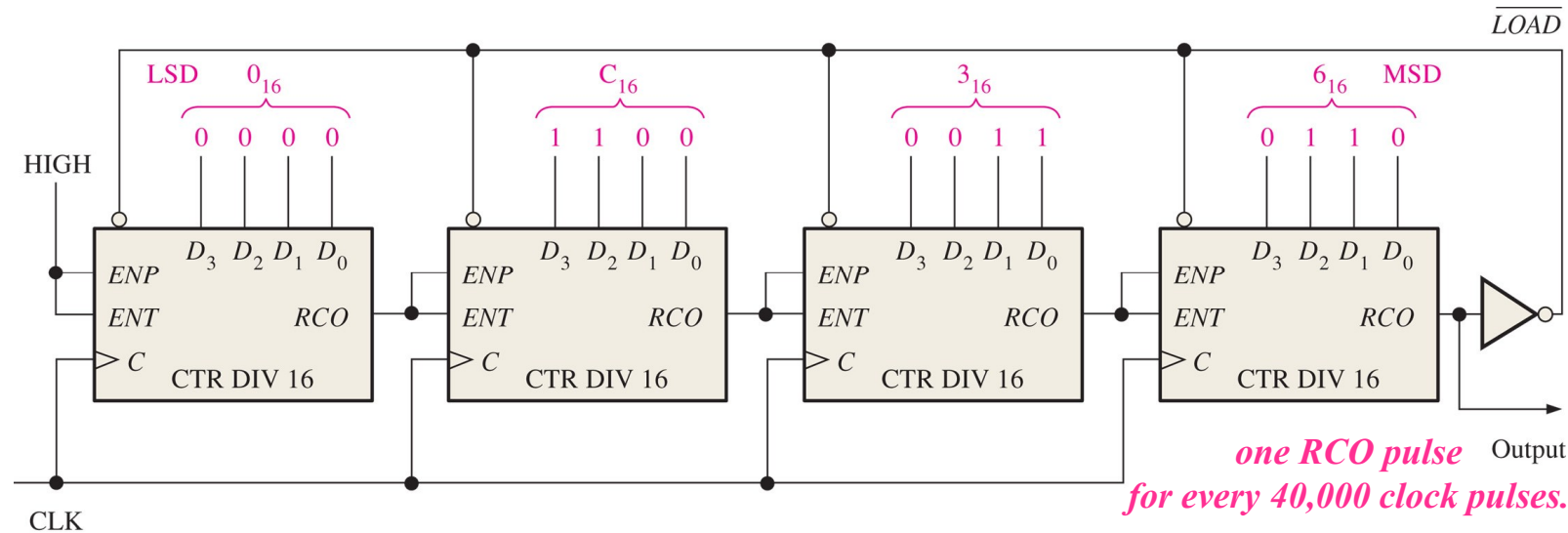
The left counter produces a terminal count (MAX/MIN) pulse for every 10 clock pulses.
The right counter produces a terminal count (MAX/MIN) pulse for every 100 clock pulses

Cascaded counters w/truncated sequences



Often an application requires an overall modulus that is less than that achieved by full-modulus cascading.

A divide-by-40,000 counter using 74HC161 4-bit synchronous binary counters. Note that each of the parallel data inputs is shown in binary order (the right-most bit D_0 is the LSB in each counter).



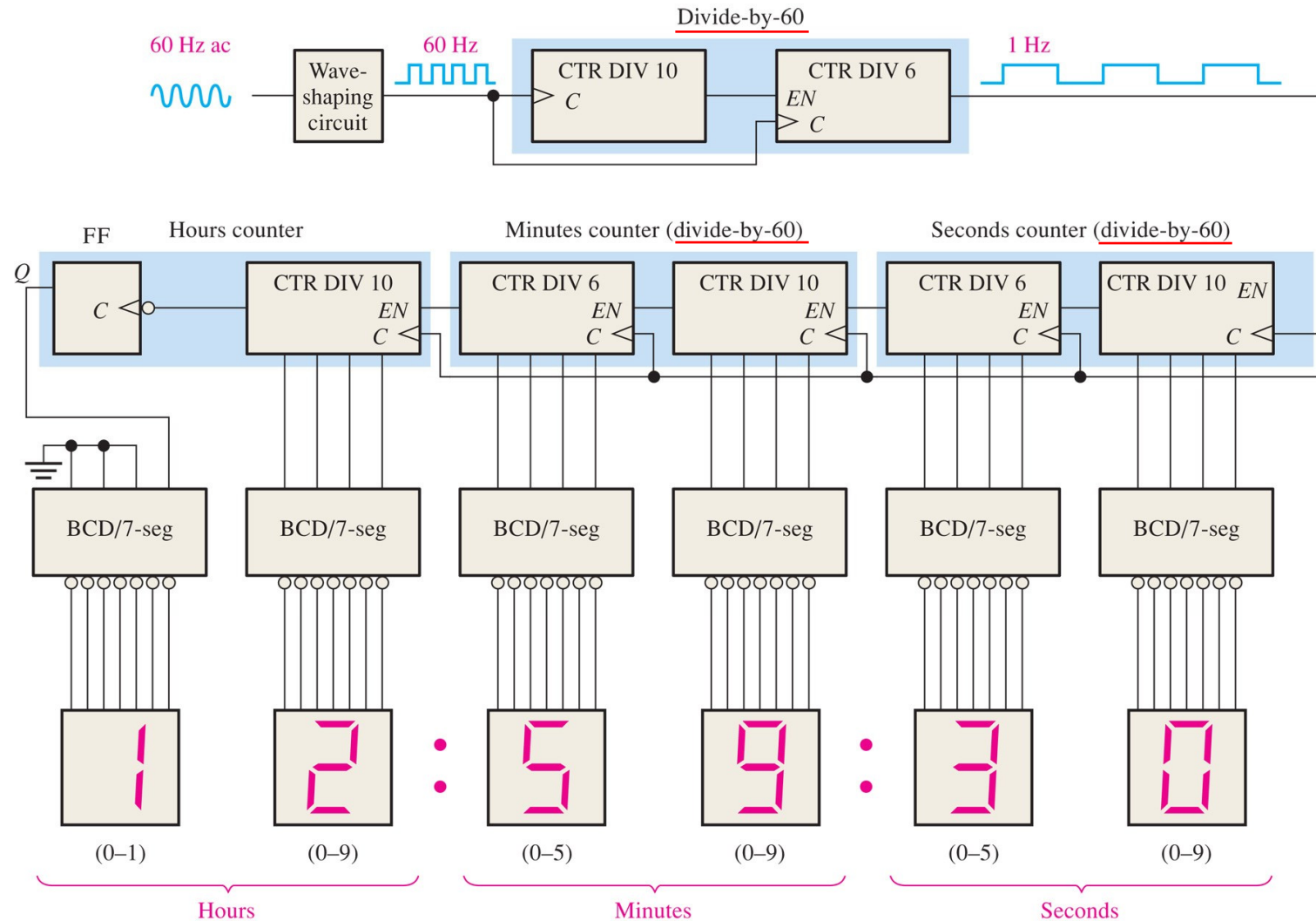
full-modulus cascading: the product of the individual moduli of all the cascaded counters.

$$2^4 2^4 2^4 2^4 = 2^{16} = 65,536 \text{ (each 4-bit counter is modulus } 2^4 = 16\text{)}$$

The counters are PRESET to $(63C0)_{16} = 25,536$ each time it recycles meaning the counter goes through $65,536 - 25,536 = 40,000$ states (modulus 40,000)

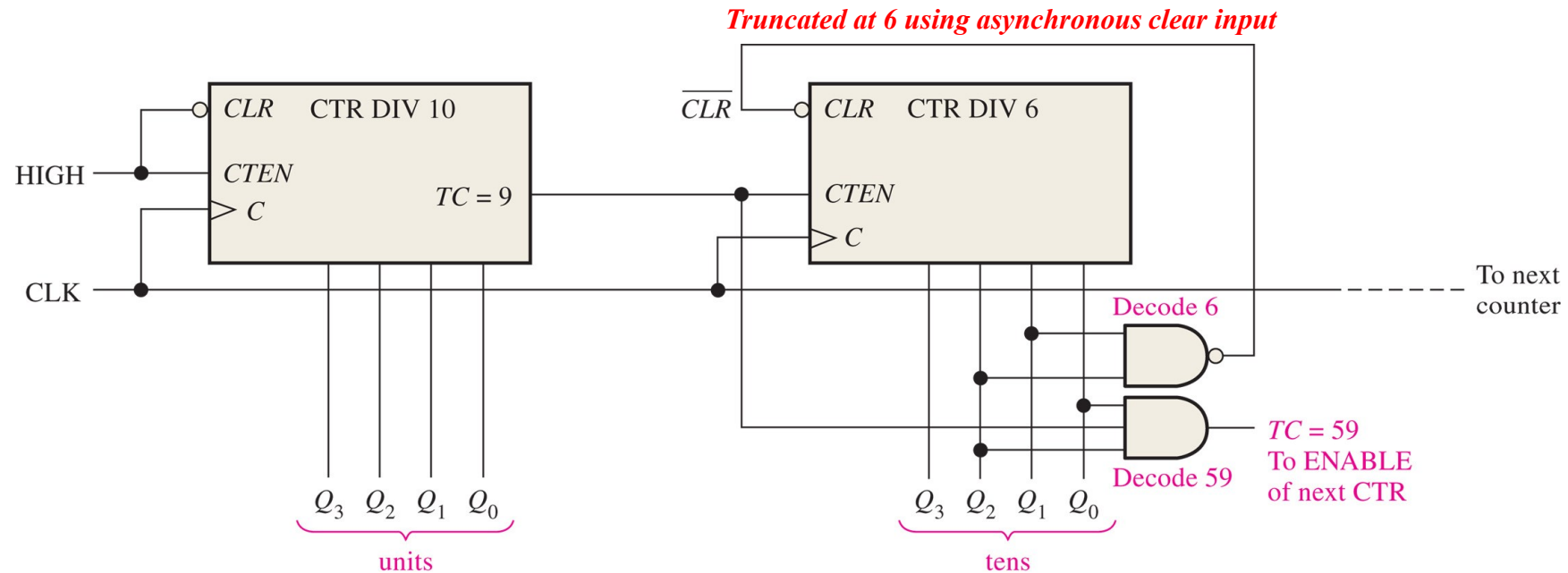
Any modulus can be achieved by synchronous loading of the counter to the appropriate initial state on each cycle.

Simplified logic diagram for a 12-hour digital clock



Divide-by-60 counter

Logic diagram of typical divide-by-60 counter using synchronous decade counters.
Note that the outputs are in binary order
(the right-most bit is the LSB).



Logic diagram for hours counter and decoders.

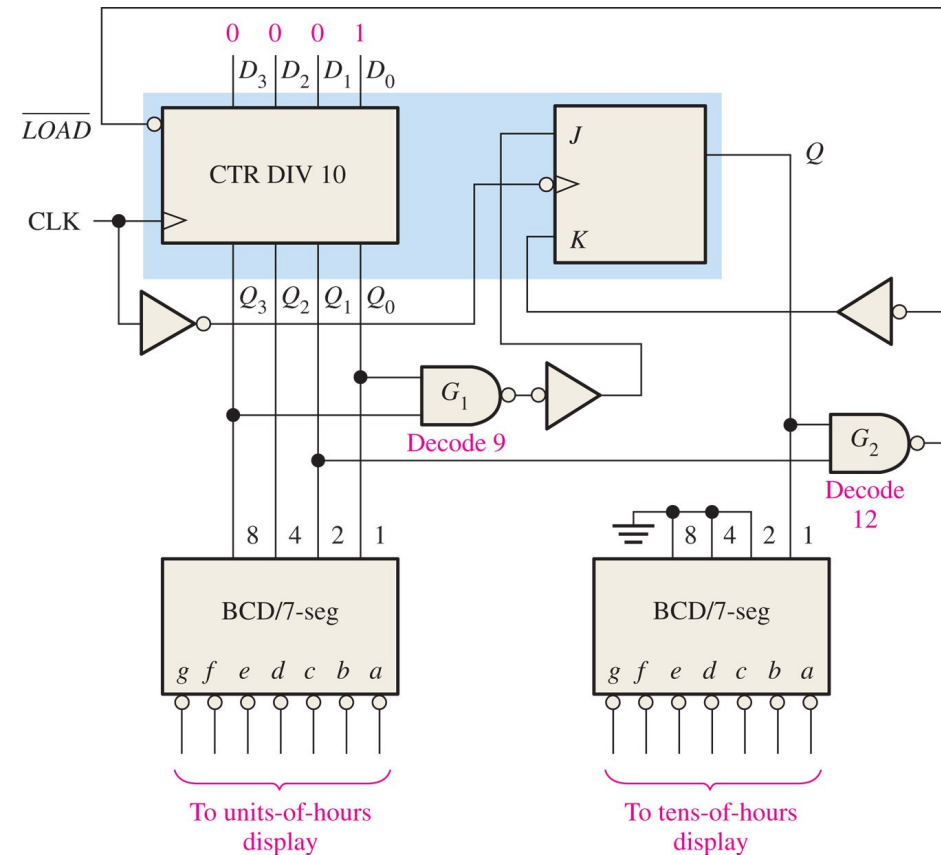
Note that on the counter inputs and outputs, the right-most bit is the LSB.

1. Initially both the decade counter and the flip-flop are RESET, and the decode-12 gate and decode-9 gate outputs are HIGH.
2. On the clock pulse that recycles decade counter from 9 back to 0, the flip-flop goes to the SET state ($J = 1$, $K = 0$).

This illuminates a 1 on the tens-of-hours display. The total count is now 10

3. In state 12 the Q_1 output of the decade counter is HIGH, the flip-flop is still SET, and thus the decode-12 gate output is LOW.

This activates the LOAD input of the decade counter. On the next clock pulse, the decade counter is preset to 0001, and the flip-flop is RESET ($J = 0$, $K = 1$). This always causes the counter to recycle from 12 back to 1 rather than back to 0.

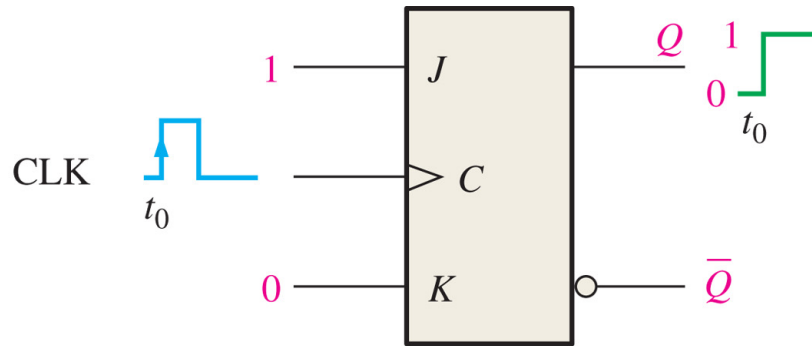


Main Content

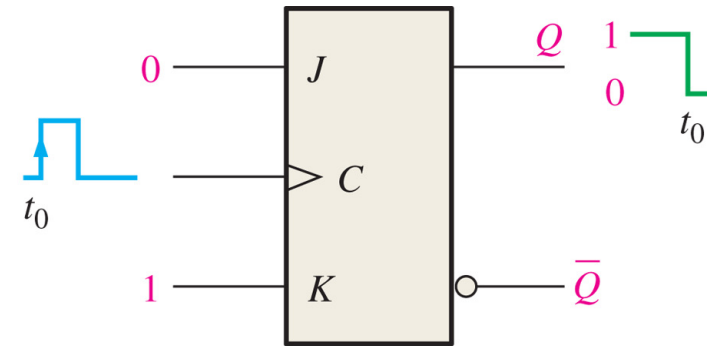
JK Flip-Flop Operation and Timing

- JK Flip-Flop Behavior (Set, Reset, Toggle)
- Timing Waveform and Characteristic Table

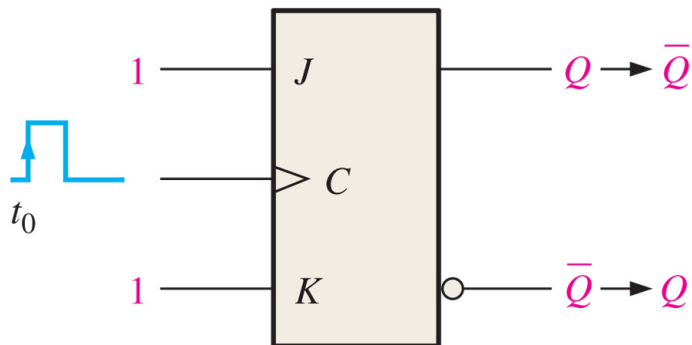
JK Flip-Flop



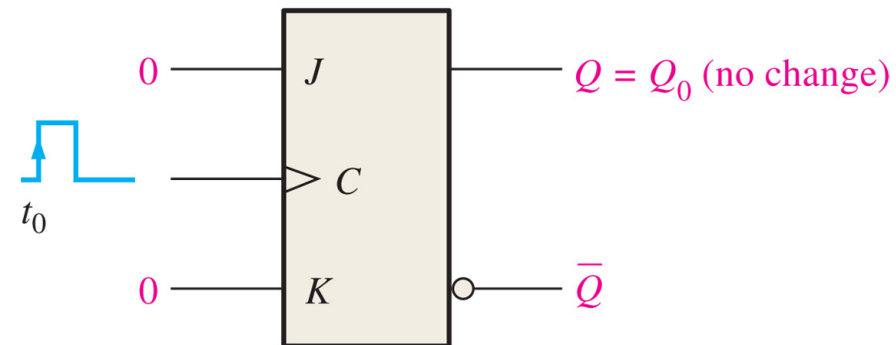
(a) $J = 1, K = 0$ flip-flop SETS on positive clock edge. (If already SET, it remains SET.)



(b) $J = 0, K = 1$ flip-flop RESETS on positive clock edge. (If already RESET, it remains RESET.)

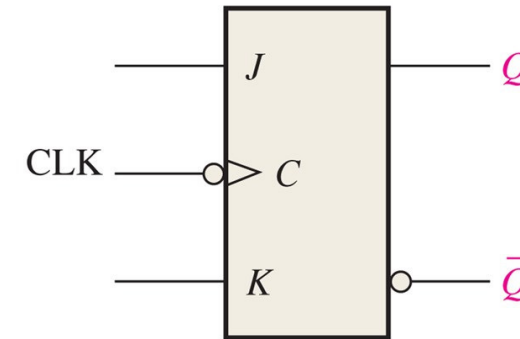
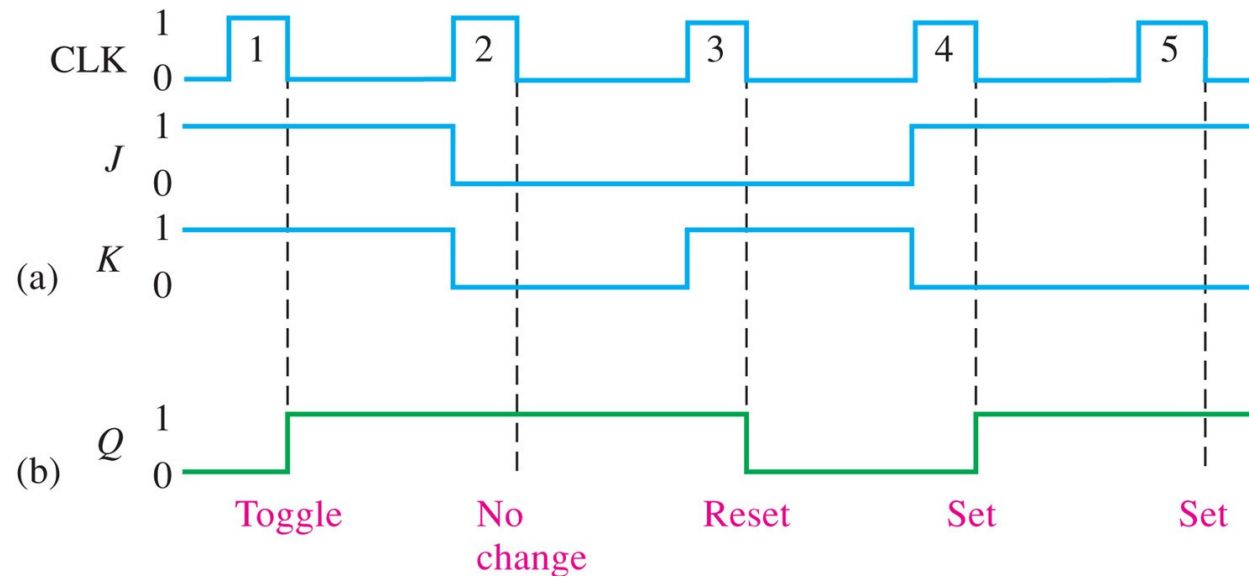


(c) $J = 1, K = 1$ flip-flop changes state (toggle).



(d) $J = 0, K = 0$ flip-flop does not change. (If SET, it remains SET; if RESET, it remains RESET.)

Timing Waveform



Characteristic Table

JK Flip-Flop			
J	K	$Q(t + 1) = Q^+$	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

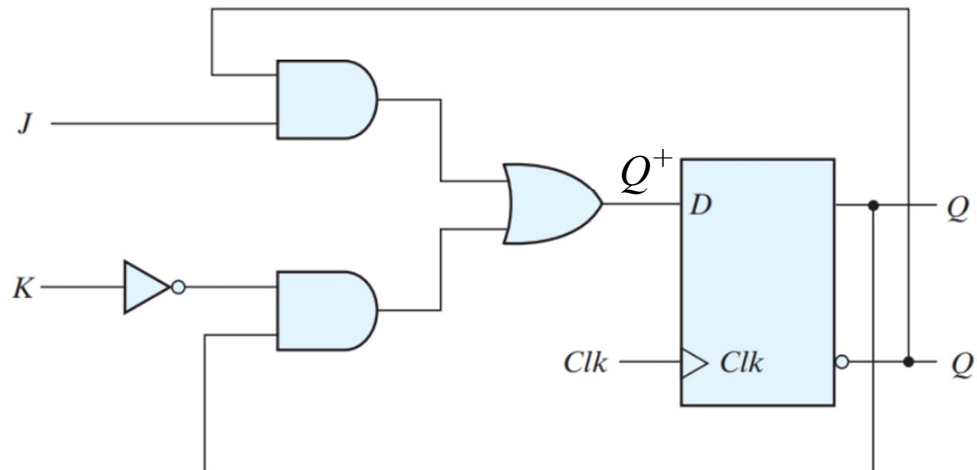
Expand into 3-input truth table where present state Q is one of the inputs and next state Q^+ is the output. And then write Q^+ using minterms or maxterms

$$Q^+ = \bar{J}\bar{K}Q + J\bar{K}\bar{Q} + J\bar{K}Q + JK\bar{Q}$$

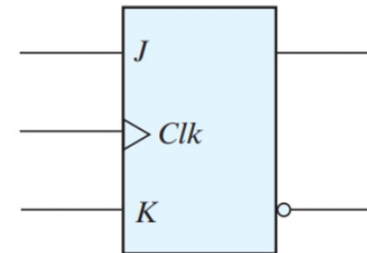
$$= J\bar{Q} + \bar{K}Q$$

J	K	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Complement also known as Toggle mode



(a) Circuit diagram



(b) Graphic symbol

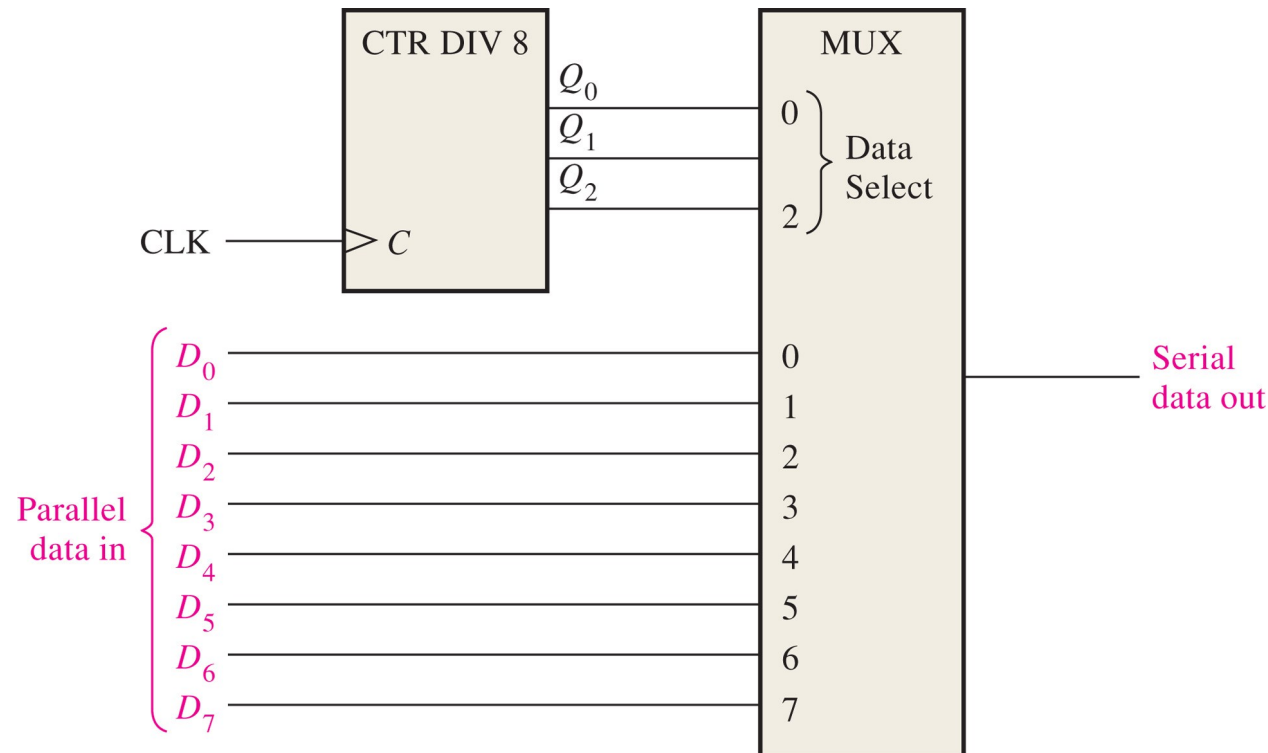
Main Content

Real-World Counter Applications

- Digital Clock Design (Divide-by-60 and Hours Counter)
- Parallel-to-Serial Data Conversion
- Automobile Parking Control System

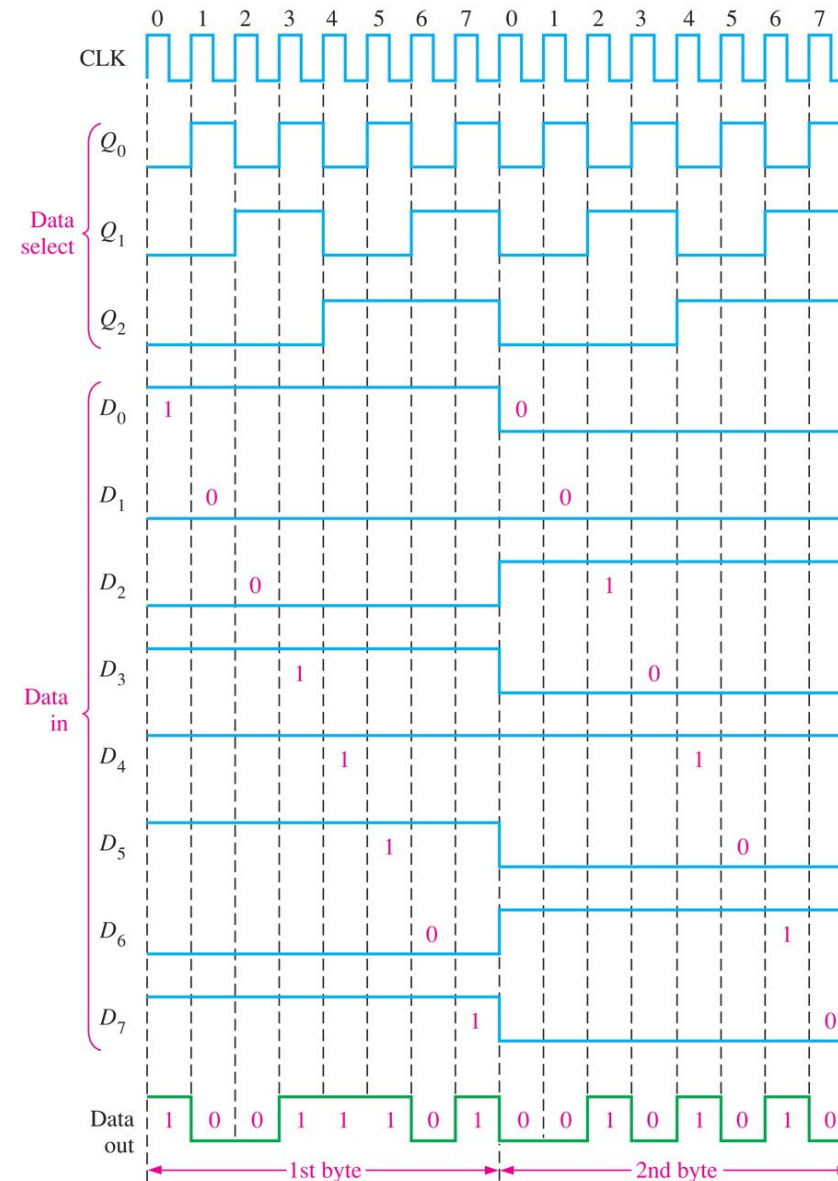
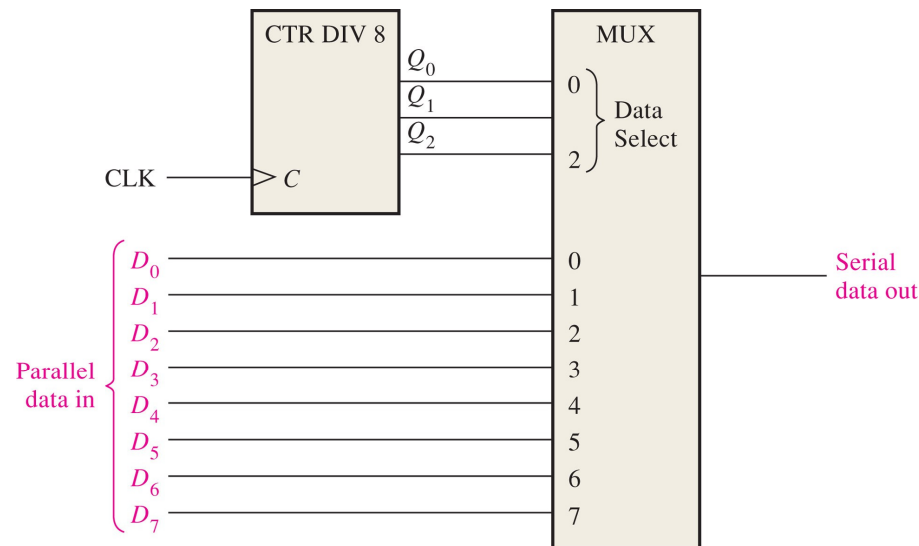
Parallel-to-serial data conversion

- The parallel data bits on the MUX inputs are converted to serial data.
- parallel data: group of bits appearing simultaneously on parallel lines
- serial data: group of bits appearing on a single line in a time sequence



Parallel-to-serial data conversion

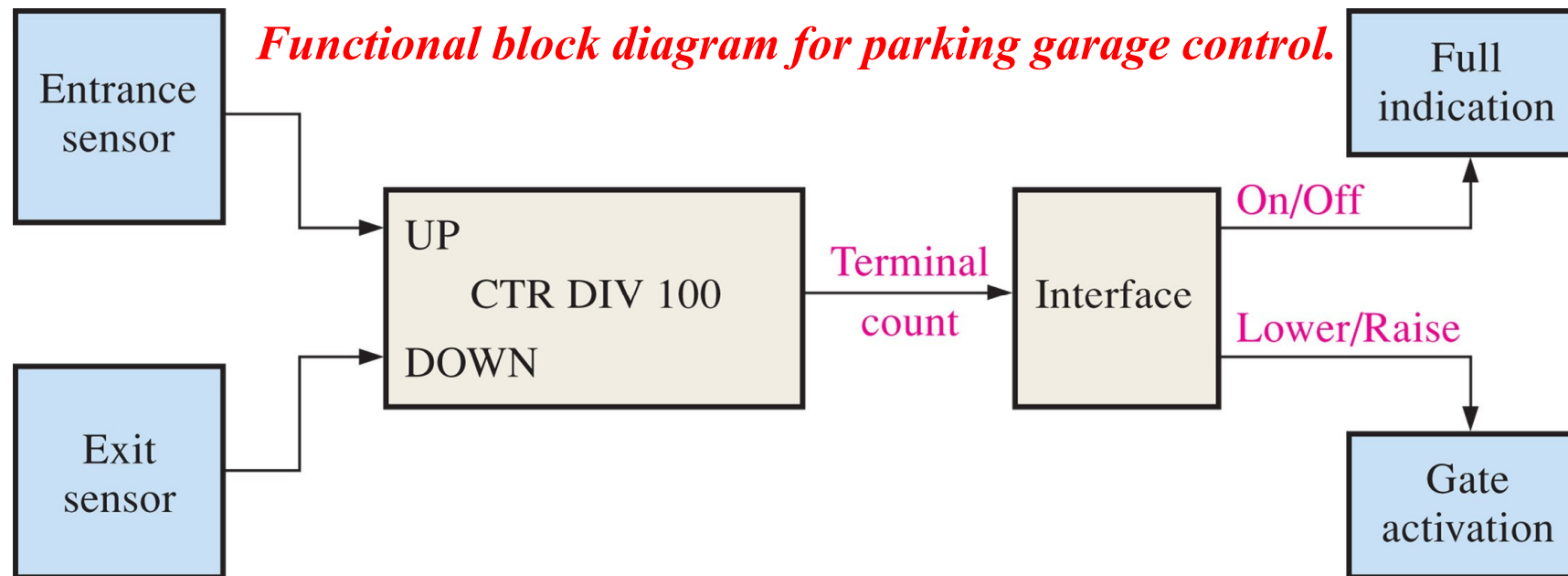
- The first byte is applied to the MUX inputs.
- As the counter goes from 0 to 7, beginning with D0, each bit is sequentially passed to the MUX output.
- After 8 clock pulses the data byte has been converted to a serial format and sent out on the transmission line.
- When the counter recycles back to 0, the next byte is applied to the data inputs



Automobile Parking Control

Design a system that would display FULL and LOWER the entrance gate bar when NO space is available in a 100-space parking garage.

- Use optoelectronic sensors at the entrance and exit of the garage
- Use an up/down counter, and an interface circuit that uses the counter output to turn the FULL sign on or off and lower or raise the gate bar at the entrance



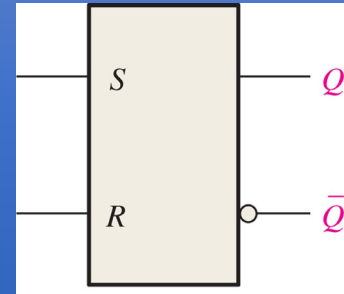
Automobile Parking Control

Each automobile entering/exiting the garage breaks a light beam, activating a sensor that produces an electrical pulse.

$\bar{Q} = 0 \rightarrow$ Counter in UP mode

$\bar{Q} = 1 \rightarrow$ Counter in DOWN mode

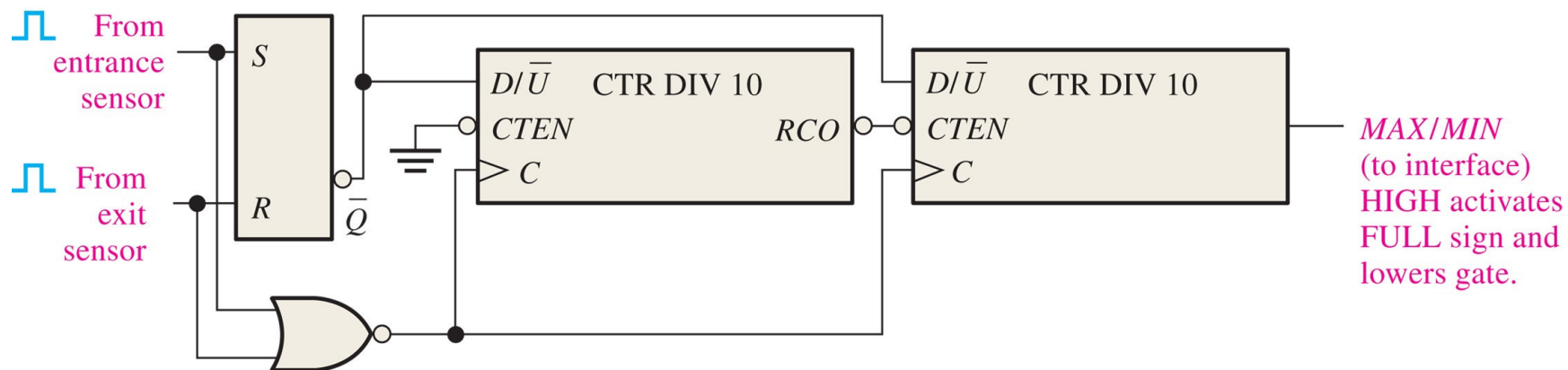
SR Latch (S = Set; R = Reset)



D	S	R	Q	State
	0	0	Previous State	No Change
0	0	1	0	Reset
1	1	0	1	Set
	1	1	?	Forbidden

SR & D Flip Flop TruthTable

Logic diagram for modulus-100 up/down counter for automobile parking control.



We will visit the topic of counters again in future lectures



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Thank you for listening !

